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# Technical, financial, and emissions analyses of solar water heating systems for supplying sustainable energy for hotels in Ghana

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ABSTRACT

Ghana has the potential to deploy solar energy technologies, with its solar irradiation varying between 4 and 6.5 kWh/m$^2$/day. However, the country's dependence on fossil fuels for generating electricity and heat remains prevalent. This study aims to assess the technical and financial feasibility of installing solar water heating (SWH) systems in hotels within Ghana. RETScreen software is used to conduct the technical, financial, and emission analyses of the SWH system for five cities in Ghana, including Accra, Cape Coast, Kumasi, Tamale, and Wa. The findings show that SWH systems are feasible for Ghanaian hotels, with solar fractions ranging from 61.2% in Kumasi to 78.5% in Wa. Also, installing the SWH system yields a positive net present value for all the cities. The implication is that installing SWH systems in hotels operating in Ghana is financially viable and attractive for investment. In addition, payback periods infer that the SWH systems can generate a return on investment in a reasonable time frame, especially considering the equity payback period. Furthermore, about 58.7 tonnes of carbon dioxide ($CO_2$) emissions could be avoided annually by installing the SWH system in the selected cities. This study's findings suggest that hotels can achieve long-term financial savings on electricity costs by utilising solar energy to heat water. This, in turn, reduces their reliance on fossil fuel consumption while actively pursuing their sustainability objectives. The study findings are crucial in assisting hotel owners in making informed decisions on SWH systems.

## Introduction

Energy consumption is a fundamental aspect of modern society and is closely linked to sustainable development [1]. The abrupt rise in energy demand due to industrialisation has led to an increase in greenhouse gas (GHG) emissions, especially carbon dioxide ($CO_2$), which significantly contributes to climate change [2]. The projected growth in world primary energy demand is expected to exacerbate this problem [3]. Developing countries are particularly vulnerable to the negative impacts of energy consumption, as they often lack access to affordable and clean energy [4,5].

Sustainable energy initiatives supporting social and economic growth are more likely to succeed. Building a more sustainable global economy could help reduce GHG emissions and support sustainable development and climate action [6]. Therefore, it is vital to explore renewable and low-carbon energy sources that can meet the growing energy needs while mitigating the adverse effects of GHG on the environment. Utilising low-carbon emissions and alternative energy sources is vital for attaining sustainable development goals [7,8]. Amongst renewable energy sources, solar energy is widely available and can help tackle the adverse impact of climate change while promoting clean energy technologies [9].

The utilisation of solar energy has emerged as a highly promising and viable substitute for conventional energy sources. This is particularly applicable in developing countries with limited access to electricity. According to Şerban et al. (2016), solar energy has economic and environmental benefits that can be used to achieve significant energy

*List of Abbreviations:* BCR, Benefit-cost ratio; $CO_2$, Carbon dioxide; CPC, Compound parabolic collectors; ECOWAS, Economic Community of West African States; ECREEE, ECOWAS Regional Centre for Renewable Energy and Energy Efficiency; EPP, Equity payback period; ETC, Evacuated tube collector; FPC, Flat plate collector; GHG, Greenhouse gas; GW, Giga Watt; IRR, Internal rate of return; L, Litre; m$^2$, Square metre; MWh, Mega Watt hour; NCF, Annual net cash flow; NPV, Net present value; SA, South Africa; SPP, Simple payback period; SWH, Solar water heating; SSA, Sub-Saharan Africa; $tCO_2$, Tonnes of carbon dioxide; yr, Year.

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savings with no direct GHG emissions. Solar energy can be utilised in its thermal form for many purposes, such as water heating for domestic, commercial, and industrial applications [11].

Şerban et al. [10] alluded to the fact that a cost-effective and widely-used method of harnessing solar energy is through solar water heating (SWH) systems. Such systems are ideal for low-temperature applications below 50 °C due to their simplicity, minimal system components, and low investment and operational expenses. SWH systems can be classified into active and passive solar water heaters [12]. Active heating systems use electrical and mechanical components that distribute water or antifreeze solutions through solar collectors to heat water. On the other hand, passive heating systems use natural heat transfer and system design to harness and retain solar heat without mechanical or electrical components. Active systems offer more control and can be more efficient, but passive systems are simpler, cheaper, and require less maintenance.

According to Weiss [13], the total cumulated solar thermal capacity in operation worldwide by the end of 2020 is estimated to be 501 GW for solar thermal heat, with an annual supply of 407 TWh. It is worth mentioning that the energy produced from solar thermal systems installed for heating in 2020 is equivalent to 43.8 million tonnes of oil and 141.3 million tonnes of $CO_2$ savings. This demonstrates the substantial role solar energy technology could play in decreasing global GHG emissions. SWH system is a mature technology utilised worldwide to provide hot water for bathing, cleaning clothes, and dishwashing [10,11]. However, the majority of solar thermal heat in operation by the end of 2020 was installed in China and Europe, with China accounting for more than 72% of the total installed capacity.

In the Sub-Saharan African (SSA) region, the total installed capacity of solar thermal heat systems as of the end of 2019 was very low compared to other regions. Only 0.4% of the total capacity is installed in SSA, including South Africa, which has the highest installed capacity of 1633 MW [13]. However, different projects are being installed in the region, including the 10,000-apartment SWH project in Osana village in Namibia [14]. In Ghana, various SWH systems are used, including direct thermosyphon systems. The Royal Senchi Resort in Akosombo is one example of a facility that utilises a SWH system with an installed capacity of 37.8 kWth. This system preheats cold water before heating it to the required temperature using a boiler and liquefied petroleum gas [15].

Ghana, located in West Africa, is presently classified as a developing country that experiences significant levels of solar radiation throughout the year. Despite this, the nation heavily depends on fossil fuels to generate electricity and heat, resulting in high energy costs and carbon emissions [16]. The hotel sector represents one of the most substantial energy consumers in the country, and the integration of SWH systems holds the potential to reduce energy costs and carbon emissions markedly [17]. However, adopting these systems in Ghana has been limited, mainly due to the high upfront costs and a lack of awareness of the benefits [18].

The Ghanaian government is committed to enhancing the cost efficiency of solar and wind energy by tackling the technological challenges, institutional barriers, and market limitations that hinder the adoption of solar and wind energy technologies [19]. Notwithstanding initiatives by the government to increase the proportion of renewables and transition to a sustainable energy mix [20], the country continues to rely heavily on fossil fuels to meet its energy demands [21]. This has resulted in several environmental, economic, and social problems, such as high energy costs, limited access to electricity, and rising GHG emissions.

In view of this, the government has policies and programmes to promote renewable energy technologies, including SWH systems, to increase energy efficiency, reduce costs, and alleviate the effects of climate change. For example, the country's renewable energy masterplan aims to deploy about 135,000 units of SWH systems by 2030 [22]. Similarly, the government has pledged that more than 50% of water heating systems would stem from using solar energy by 2050 in its present national energy transition framework [23]. These initiatives demonstrate Ghana's commitment to addressing climate change and its adverse impact on socio-economic growth.

*Literature review*

Several authors have investigated the viability of adopting SWH systems for large-scale to small-scale household, industrial, and commercial applications. For example, Naveed Arif et al. [24] employed T*SOL software to investigate the feasibility of adopting a SWH system to achieve sustainable tourism in Pakistan. The authors revealed that evacuated tube collectors (ETC) showed higher solar fraction, efficiency, and $CO_2$ emissions savings than flat plate collectors (FPC) and unglazed collectors. Also, investing in the SWH system in Pakistan has a payback period between 4.4 and 6.6 years. Similarly, Sinethemba et al. [25] examined the economic and environmental potential of substituting electric water heaters with SWH in Dimbaza, South Africa. The findings show that the simple payback period is about 5.02 years. In addition, deploying SWH systems in Dimbaza's households could be feasible from a technological, economic, and ecological standpoint due to the possible energy savings and reduced GHG emissions.

Likewise, Ibrahim et al. [26] applied T*SOL simulation software to assess the viability of deploying SWH systems for swimming pools in tourist hotels and resorts in El Gouna City, located in Egypt. The findings indicate that installing the SWH systems saves about 6720 kg of $CO_2$ annually, saving USD 6102. Also, the payback period for the SWH system in the context of swimming pools is approximately 14 months. Nevertheless, choosing the SWH system in hotels and resorts is an exemplary and highly recommended alternative. In addition, Azad Gilani & Hoseinzadeh [27] numerically investigated the impact of utilising a compound parabolic collector (CPC) compared to a standard FPC. The study considered multiple locations between 0 and 60° north latitude. CPC was observed to have lower auxiliary power consumption than FPC. In addition, the locations with the maximum solar radiation availability demonstrated enhanced CPC energy savings.

Correspondingly, Singh et al. [9] used RETScreen software to study the environmental, economic, and technical performance of domestic SWH systems in India. The findings indicate that the SWH project could save 739 $tCO_2$ of GHG emissions from the household sector over the 25 years of the system's lifetime. Also, the SWH systems have an average payback period of 7 to 10 years, depending on the site. Another study by Singh et al. [11] showed that using SWH systems could result in GHG savings of 155.9 $tCO_2$, 2419.3 $tCO_2$, and 492.3 $tCO_2$, respectively, for residential, commercial, and industrial sectors. Also, the average simple payback period is about 5 years for residential, 9 years for commercial, and 4 years for the industry sector. Correspondingly, Cruz et al. [28] alluded to the fact that SWH is economically feasible for the household sector in Brazil.

Raza et al. [29] used RETScreen software to evaluate the efficiency of SWH system in seven cities in India. The authors found that the system payback period for the seven cities ranges from 5 to 15 years. Moreover, Gujrat is the best city for installing SWH systems. Dioha et al. [30] also applied RETScreen software to evaluate the technical and financial viability of SWH systems in Nigeria's household sector. Their results showed that a maximum solar fraction of 69% could be achieved using a single evacuated SWH, with major environmental and economic benefits. Idowu et al. [31] performed a techno-economic pre-feasibility study of SWH in apartment buildings in South Africa and found that a SWH system with a 32% solar fraction and a 3.05 benefits-to-cost ratio is realisable with positive overall outcomes in Cape Town city.

Fayaz et al. [32] applied F-Chart to perform a thermal and economic analysis of ETC SWH system in Malaysia. It was observed that the ETC SWH system attains a fraction of 1 to fulfil the hot water demand of 150 L/day. Rajab et al. [33] also confirmed the economic and environmental advantages of installing SWH systems in Libya. Similarly, Hoseini et al. [34] found that SWH system is economically viable and provides a promising avenue for mitigating GHG emissions within the residential sector of Iran. Furthermore, Joubert et al. [35] revealed that current solar thermal systems in South Africa could already compete with heat sources such as electricity and gas, even though low fuel prices in the country are a barrier to developing SWH systems.

Almahmoud et al. [36] examined the performance of SWH systems for hospitals in Saudi Arabia. The authors found that using glazed FPC outperforms ETC due to its shorter payback periods and higher annual life cycle savings. Similarly, Abd-ur-rehman & Al-sulaiman [37] employed RETScreen software to analyse evacuated and glazed collectors SWH systems for households in five different cities of Saudi Arabia. The findings justified the reliability of evacuated SWH collectors in obtaining a higher solar fraction and reducing GHG emissions. Correspondingly, Soomro et al. [38] employed RETScreen software to save energy and $CO_2$ emissions by installing a SWH system in Pakistan. The results indicate that SWH application in the domestic sector is vital for the energy transition in Pakistan and could significantly help to save fuel while reducing GHG emissions. Furthermore, Håkansson et al. [39] assessed the performance of SWH systems in Algeria. The results showed potential energy savings by yielding more than 905 $tCO_2$ of GHG reduction per annum. The authors further pointed out that the low fuel costs in Algeria represent one of the barriers to developing SWH technologies.

The literature mentioned above shows that SWH systems have garnered significant interest as an effective means to meet increasing energy needs in the residential, commercial, and industrial sectors. It is worth noting that prior studies mainly focused on the system's technical indicators (sizing, design, and performance assessments), financial feasibility, and emission savings potential. However, one crucial factor overlooked in previous research is the correlation between solar irradiation and the heat delivered by the SWH systems. The correlation between solar irradiation and heat delivery holds significant importance in determining the overall performance and reliability of SWH systems.

In view of this, assessing this correlation can provide insights into how effectively solar irradiation translates into useful thermal energy, providing a crucial benchmark for system optimisation. Therefore, this study introduces a Pearson correlation test between solar irradiation and the heat delivered by SWH systems, ensuring a robust evaluation of their performance. In addition, the study also applied the P-value to assess the strength of the correlation between solar irradiation and the heat delivered, which has been overlooked in previous studies. This aims to provide helpful insight into the practical deployment of SWH systems, leading to higher energy savings, cost savings, and GHG emissions reduction in the hospitality sector.

The study's main objective is to conduct technical, economic, and emissions analyses of SWH systems for hotels operating in Ghana. Likewise, the study identifies the potential barriers that could hinder stakeholders' adoption of SWH systems. The study findings are crucial in assisting hotel owners in making informed decisions regarding adopting SWH systems. Additionally, this study could contribute to developing sustainable energy policies to promote renewable energy sources in Ghana. The study's remaining sections are as follows: Section 2 presents the methods and materials employed to achieve the study's goal. Also, Section 3 highlights the results and discussions, and Section 4 provides a conclusion and remarks on the study's key findings.

## Methodology

This section presents the methods and materials used to conduct the technical, financial, and emission analyses of the SWH system.

### *RETScreen expert solar water heating system model*

RETScreen expert software 8.1 is utilised to conduct the technical, financial, and emissions analyses of the SWH system for Ghanaian hotels. RETScreen is a widely used tool for the feasibility and performance analysis of different energy systems [30,40]. The software allows users to select the appropriate location and enter data to calculate the annual energy yield. The SWH system model is suitable for performing analysis for domestic hot water applications, hotel and motel water heating technology analysis, and outdoor and indoor swimming pools in commercial, residential, and institutional buildings [41]. Fig. 1 displays the SWH system model adopted in RETScreen software. The software has an optimised algorithm that allows it to perform detailed financial analysis, GHG emissions reduction scenarios, as well as risk and sensitivity analysis [40].

**Table 1**
Selected cities for assessing the viability of installing SWH systems.

| Region | Latitude (°N) | Longitude (°E) | Elevation (m) |
|---|---|---|---|
| Accra | 5.6 | −0.2 | 68 |
| Cape Coast | 5.1 | −1.3 | 60.5 |
| Kumasi | 6.7 | −1.6 | 287 |
| Tamale | 9.5 | −0.9 | 168 |
| Wa | 10.1 | −2.5 | 271 |

**Table 2**
Technical specifications for selected SWH for selected cities [30].

| Parameter | Value |
|---|---|
| Manufacture | Ritter Solar |
| Model | CPC 12 INOX |
| Gross area per solar collector | 2.28 $m^2$ |
| Aperture area per solar collector | 2.01 $m^2$ |
| $F_R(\tau\alpha)$ coefficient | 0.56 |
| $F_R U_L$ coefficient | 1 $(W/m^2)/°C$ |
| Temperature coefficient | 0 $(W/m^2)/°C$ |

### *Study location and climate data*

Ghana is a country in West Africa that shares borders with Togo to the east, Burkina Faso to the north, Côte d'Ivoire to the west, and the Gulf of Guinea to the south [42]. The location of Ghana in the context of Africa is shown in Fig. 2. The country's strategic location near the equator provides solar irradiation, between 4 and 6.5 $kWh/m^2/day$, making it ideal for harnessing solar energy [43]. The country's sunshine hours, between 1800 and 3000 hours per annum, provide significant potential for deploying solar energy technologies [44,22].

This study selected five cities, including Accra, Cape Coast, Kumasi, Tamale, and Wa, to assess the feasibility of installing SWH systems. The cities' latitude, longitude, and elevation are highlighted in Table 1. The cities were selected due to the availability of data. Fig. 3 shows the average global horizontal solar irradiation for the selected cities. It can be observed in Fig. 3 that the cites daily solar irradiation ranges from 3.35 to 6.15 $kWh/m^2/day$. Also, it is worth mentioning that global daily solar irradiation in Ghana increases gradually from the south to the north, with high solar irradiation between January and May (Fig. 3).

### *Solar collector technical specification and cost*

Ghana has several hotels throughout the country. These hotels are categorised under non-star, 1-star, 2-star, and 5-star hotels in Accra, Kumasi, Takoradi, and other cities for business people, travellers, tourists, and hospitality. However, a typical 3-star hotel is considered in this study. According to a survey by Danso et al. [46], there were 40 3-star hotels in Ghana in 2018, with the majority having more than 50 rooms. In the same year, the occupancy rate of a 3-star hotel was about 54% in Ghana. This rate increases to 59.2% in 2019 and decreases to 31% and 18%, respectively, in 2020 and 2021 [47]. In view of this, this study considers a 3-star hotel with 70 rooms and an average occupancy rate of 60%.

Table 2 presents the selected solar collector's technical parameters and specifications. This collector is selected because it is appropriate for pitch, flat roofs, free-standing mounting, and façade assembly. In addition, it yields higher energy for a small collector gross area [48]. Other technical input data required to size the SWH system and perform

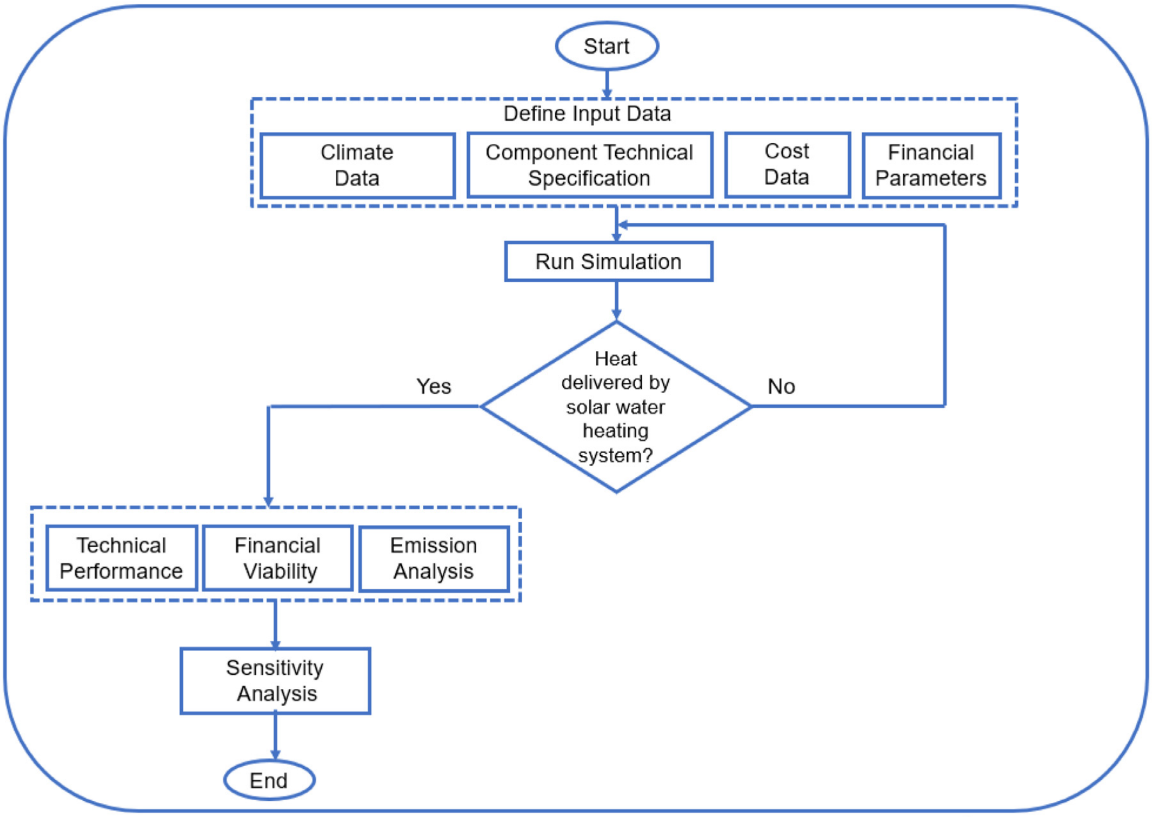


**Fig. 1.** Framework of SWH system model in RETScreen software [40].

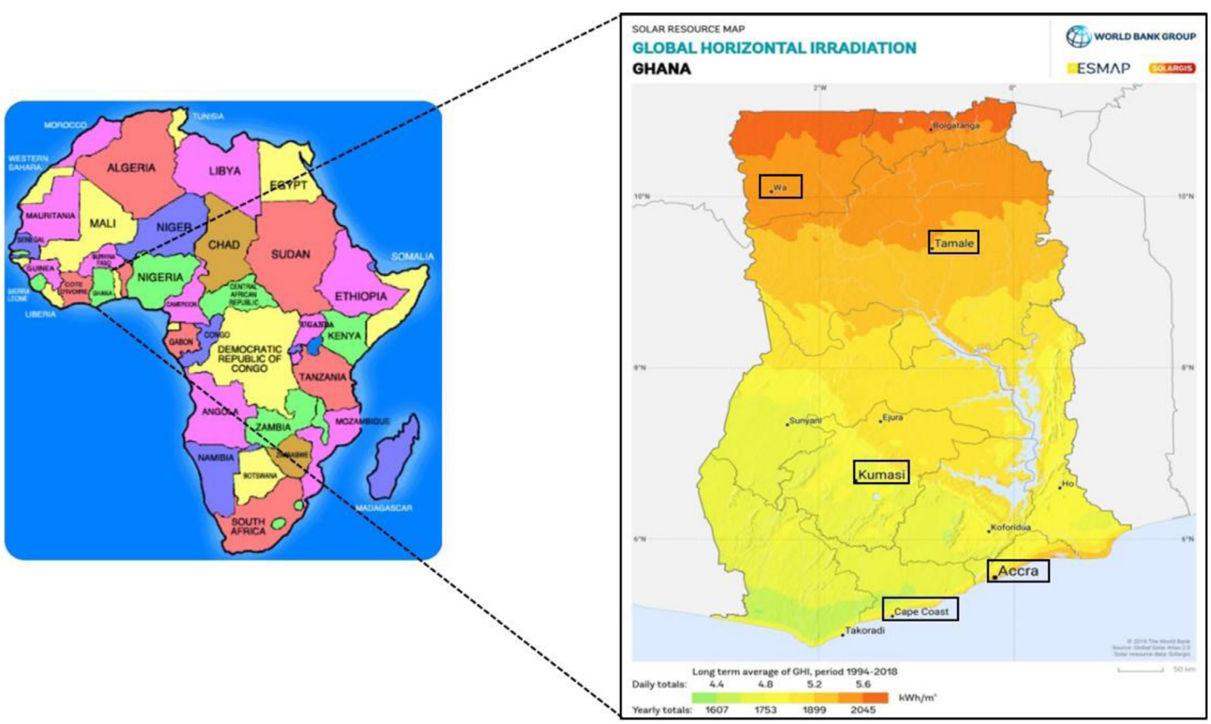


**Fig. 2.** Ghana's location in the context of Africa (left), and global horizontal irradiation potential for chosen cities (right) [45].

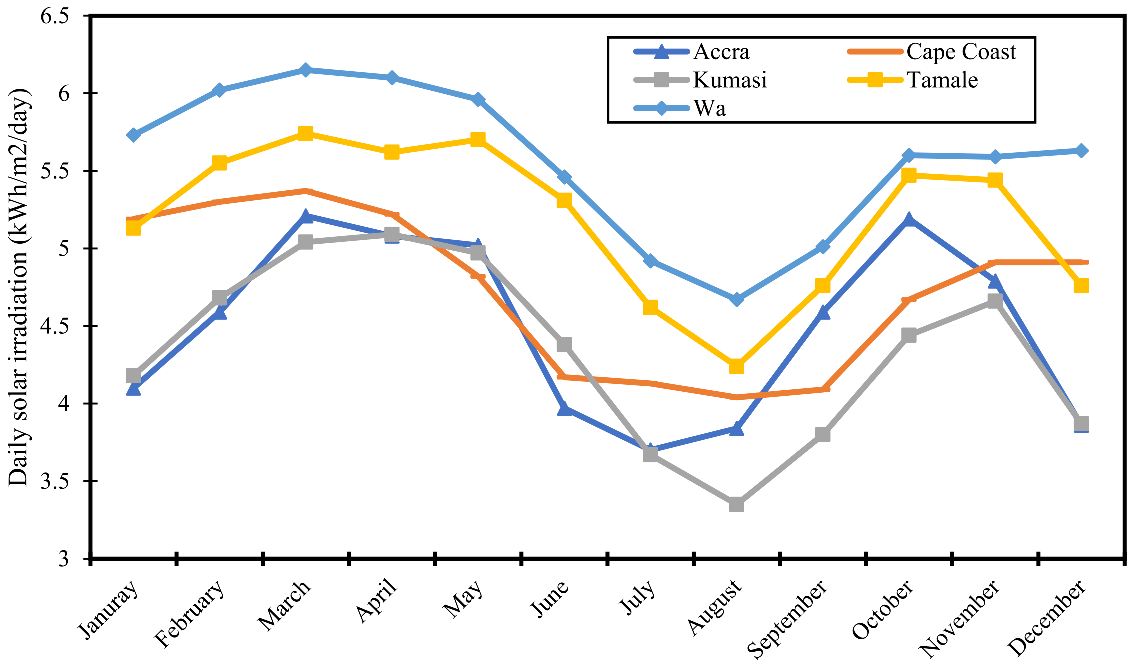


**Fig. 3.** Daily horizontal solar irradiation for different cities in Ghana.

**Table 3**
Summary of input parameters to be used to assess the feasibility of the SWH system.

| Parameter | Value | Source/Remarks |
|---|---|---|
| Average number of rooms | 70 | Deduce from a typical 3-star hotel in Ghana |
| Occupancy rate (%) | 60 | Deduced from a typical 3-star hotel in Ghana |
| Hot water temperature ( °C) | 60 | A typical value of 60 °C or higher is used for domestic or industrial applications. |
| Solar tracking mode | No | The solar collector would be installed on a fixed structure. |
| Slope/Tilt angle | 5.1° – 10.1° | The slope/tilt angle is equal to the latitude of each city. The latitude varies from 5.1° – 10.1° |
| Azimuth | 0° | This is the preferred collectors' orientation due to the study location. |
| Operational time (h) | 24 | Typical operation time for Ghanaian hotels. |
| Solar water heating type | Evacuated | Evacuated SWH systems exhibits better performance under various climatic conditions [51] |
| Inflation rate (%) | 10 | [52] |
| Discount rate (%) | 10 | [52] |
| Miscellaneous (%) | 2 | Authors' assumptions based on similar projects [52] |
| Total Initial cost (USD) | 17,200 | This is the total cost of installing the SWH system with a 200 L storage capacity [50]. This was estimated using an exchange rate of GH₵10.95 = 1 USD from the Bank of Ghana as of 13 April 2023). |
| Operation and maintenance cost (USD) | 150 | Authors' assumptions based on personal interaction with local SWH system installers in Ghana. |
| Storage | Yes | Systems with storage should be explored for domestic hot water or commercial uses where the SWH system must deliver a considerable portion of the water heating load. |
| Storage capacity (L/m$^2$) | 74.7 | Computed based on storage capacity and solar collector area. |
| Transmission losses (%) | 5 | [53] |
| GHG emission factor (tCO2) | 0.36 | [53] |
| Fuel cost escalation rate (%) | 2 | This is the project's expected annual average fuel cost increase. Assumed from [52]. |
| Reinvestment rate (%) | 10 | Typical reinvestment rates vary from 1 to 18%. |

financial viability is highlighted in Table 3. About 3003 L of hot water could be stored in a storage tank based on the collector area.

Currently, 150 L of SWH storage capacity in the Ghanaian market costs approximately GH₵ 6900 without installation costs [49]. In addition, the total cost of installing SWH systems with a 200 L storage capacity is about GH₵ 12,500 [50]. This implies that about USD 1142 (using an exchange rate of GH₵10.95 = 1 USD from the Bank of Ghana as of April 13, 2023) is required to install the 200 L SWH. In this context, the initial installation cost for the estimated SWH with a 3003 L storage capacity in this study will theoretically be around USD 17,200.

*Governing equations adopted in RETScreen expert swh system model*

*Technical analysis*. RETScreen calculates the energy demand for heating water using Eq. (1) [11,28]:

$$Q_{load} = C_p \rho V_l (T_h - T_c) \tag{1}$$

$Q_{load}$ is the required energy for water heating, $V_l$ is the water quantity, $T_h$ is the hot water temperature, $T_c$ is the cold water temperature, and $C_p$ and $\rho$ represents the specific heat capacity of water and density of water, respectively.

For a glazed or evacuated tube collector, the energy received by the solar collector per unit collector area and per unit time is estimated as follows [11,54]:

$$\dot{Q}_{coll} = F_R(\tau\alpha)G - F_R U_L \Delta T \tag{2}$$

$F_R$ is the collector heat removal factor, $\tau$ is the cover transmittance, $\alpha$ is the absorber absorptivity, G is the global incident solar radiation on the collector, $U_L$ is the overall heat loss, while $\Delta T$ is the temperature difference.

During the water heating process, some of the energy that heats the water is provided by solar thermal energy, while the remaining is received through a backup system [28]. This is characterised by the solar fraction, which represents the total heating load delivered by the solar energy and is expressed by Eq. (3):

$$S = \frac{L - L_A}{L} = \frac{L_S}{L} \tag{3}$$

With L is the amount of energy from the conventional system, $L_A$ is the backup energy supplied by the SWH system and $L_S$ is the solar energy supplied by the SWH system.

*Financial indicators.* This study employs commonly used financial indicators, including net present value, simple payback period, internal rate of return, benefit-cost ratio, and equity payback period, to assess the SWH system's financial viability for the selected cities. The following sections provide detailed descriptions and the governing equations of the financial indicators.

*Net present value (NPV).* NPV represents the difference between the present value of projects' cash inflows and outflows [40]. Positive NPVs suggest financially viable projects, and the magnitude of the NPV shows the degree to which the system efficiently utilises resources to generate income [55]. Eq. (4) is applied to compute the NPV [40,56]:

$$\mathrm{NPV} = \sum_{i=0}^{n} \frac{\mathrm{NCF_t}}{(1+i)^t} = -\mathrm{C_o} + \sum_{i=1}^{n} \frac{\mathrm{NCF_t}}{(1+i)^t} \tag{4}$$

$C_o$ is the investment cost, n is the project's economic life, $NCF_t$ is the annual net cash flow for the year t, and i represents the discount rate [55].

*Internal rate of return (IRR).* IRR represents the discount rate at which the NPV is equal to zero. It is the discount/compound rate at which the present value of returns equals that of costs [57]. The IRR is determined as follows:

$$0 = \mathrm{NPV} = \sum_{i=0}^{n} \frac{\mathrm{NCF_t}}{(1+\mathrm{IRR})^t} = -\mathrm{C_o} + \sum_{i=1}^{n} \frac{\mathrm{NCF_t}}{(1+\mathrm{IRR})^t} = 0 \tag{5}$$

*Simple payback period (SPP).* The SPP term refers to the time required for the SWH system's initial investment to be recouped in the form of benefits [55], and it is computed using Eq. (6):

$$\mathrm{SPP} = \frac{\mathrm{C_o}}{\mathrm{NCF_t}} \tag{6}$$

*Benefit-cost ratio (BCR).* The BCR measures the benefits per unit of cost investment, and if its value is greater than 1 (BCR>1), the project is viable. Eq. (7) is used to estimate the BCR of the SWH system [57]:

$$\mathrm{BCR} = \frac{\sum_{t=1}^{n} \frac{R_t}{(1+i)^t}}{\sum_{t=1}^{n} \frac{C_t}{(1+i)^t}} \tag{7}$$

$R_t$ is the return in year t, $C_t$ are the costs in year t, and n is the project lifetime.

*Equity payback period (EPP).* Equity payback is the time it takes an investor to recover its equity from project cash flows. Equity payback is a more vital indicator than simple payback because it considers project cash flows from inception and leverage (debt) [58]. Eq. (8) is applied to estimate the equity payback period:

$$\mathrm{EPP} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflows from the Investment}} \tag{8}$$

*Greenhouse gas emissions and savings.* RETScreen SWH systems model computes GHG emissions using Eq. (9) [52]

$$\mathrm{Emissions_{GHG}} = \mathrm{ES_{GHG}} * \mathrm{H_{deliverd}} * \gamma_p \tag{9}$$

Where $Emissions_{GHG}$ represents GHG that the SWH system emits, $ES_{GHG}$ represents the SWH system GHG emission factor, $H_{deliverd}$ represents the amount of heat that the SWH system delivers and $\gamma_p$ represents the SWH system losses. The annual GHG emission savings are determined as follows [52]:

$$\mathrm{Emission_{savings}} = (\mathrm{EG_{GHG}} - \mathrm{ES_{GHG}})\mathrm{H_{deliverd}}(1-\gamma_p) \tag{10}$$

where $Emission_{savings}$ represents annual GHG emission savings, and $EG_{GHG}$ represents grid GHG emission factor.

**Table 4**
SWH system heat delivered against solar irradiation.

| Region | Person Coefficient* | p-value |
|---|---|---|
| Accra | 0.969 | 0.00 |
| Cape Coast | 0.949 | 0.00 |
| Kumasi | 0.966 | 0.00 |
| Tamale | 0.937 | 0.00 |
| Wa | 0.942 | 0.00 |

*Correlation is significant at the 0.01 level (2-tailed).

*Pearson correlation coefficient test*

As indicated earlier, this study also explores the correlation between solar irradiation and the heat delivered by the SWH system. Eq. (11) is applied to determine the correlation between solar irradiation and heat delivered [59]:

$$r = \frac{n\sum ab - \left(\sum a\right)(b)}{\sqrt{\left[n\sum a^2 - \left(\sum a^2\right)\right]\left[n\sum b^2 - \left(\sum b^2\right)\right]}} \tag{11}$$

where r is the Pearson correlation coefficient, n is the number of observations in the dataset, a is the solar irradiation values, b is the heat delivered values, and Σ is the summation symbol.

## Result and discussion

The study's findings are presented and discussed in this section. It highlights the SWH system's technical performance, GHG emission savings, and the financial analysis of the SWH system for each city.

### SWH system performance

Fig. 4 shows each city's monthly heat delivered by the SWH system. It can be observed that the heat delivered by the SWH system corresponds to that of solar irradiation. The SWH system installation in the northern region delivers the highest amount of heat energy for producing hot water for bathing. The reason is that the country's global horizontal irradiation increases from the southern to the northern regions. Nevertheless, the variations in energy delivered by the SWH system installed in each city are nearly the same. For instance, the difference between the highest (Wa) and lowest (Kumasi) heat delivered is about 21%. The implication is that each city has a significant potential for installing SWH systems for generating energy for sustainable business activities.

It is worth mentioning that the heat delivered by the SWH system depends on solar irradiation. Consequently, the study investigated the strength and direction of the linear relationship between the SWH system heat delivered against the solar irradiation for the different cities, as shown in Table 4. The high correlation coefficients (ranging from 0.937 to 0.969) shown in Table 4 indicate a strong positive relationship between SWH heat delivered and solar irradiation for all the cities analysed (Accra, Cape Coast, Kumasi, Tamale, and Wa). This means that as the solar irradiation increases, there is a corresponding increase in the SWH system heat delivered. Furthermore, the p-values reported as 0.00 indicate that the observed correlations are statistically significant. In this case, the significance level is 0.01 (1%), suggesting a very low probability (less than 1%) that the observed correlations occurred by chance.

These findings suggest a strong positive relationship between solar irradiation and SWH system heat delivered in Accra, Cape Coast, Kumasi, Tamale, and Wa cities. It is worth noting that a weak correlation indicates the need for system adjustments, such as improving insulation, increasing surface area, or optimising the positioning of solar collectors to maximise solar irradiation and, consequently, heat output. Based on these results, increasing the amount of solar irradiation can lead to higher heat delivery in these cities. The results provide evidence

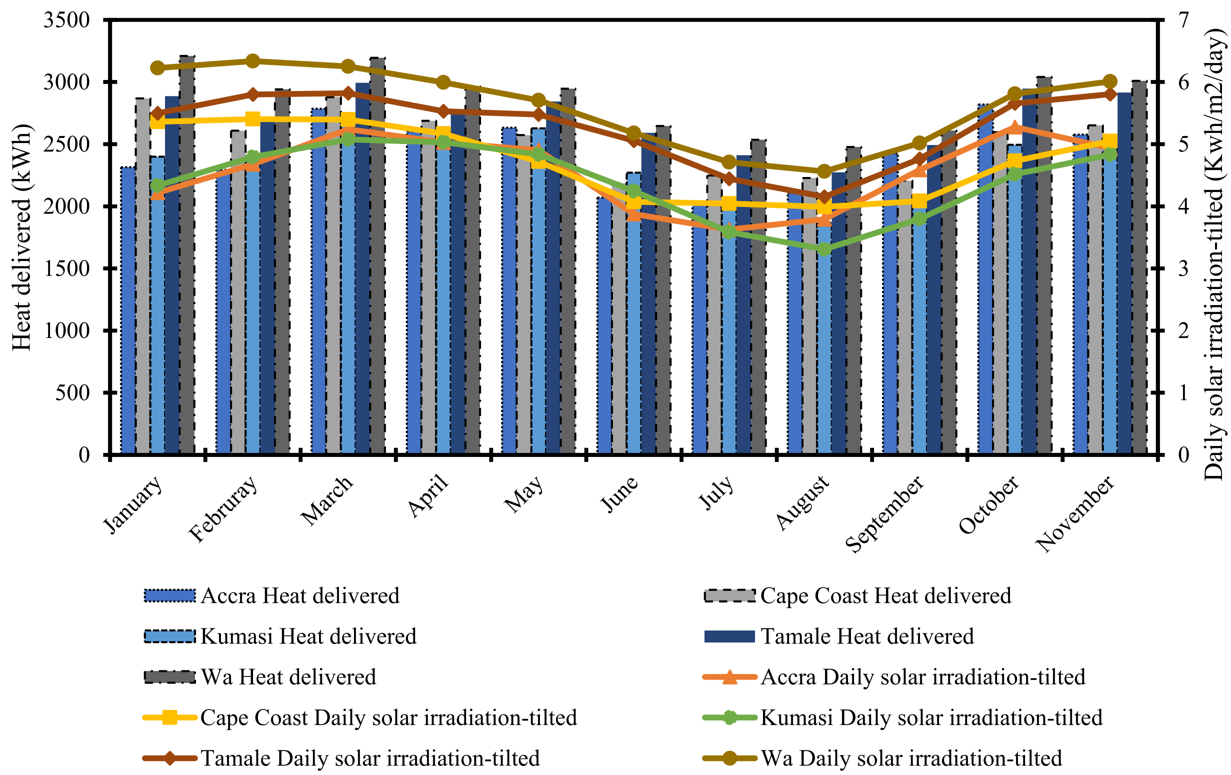


**Fig. 4.** Solar irradiation and heat delivered by the SWH for each city.

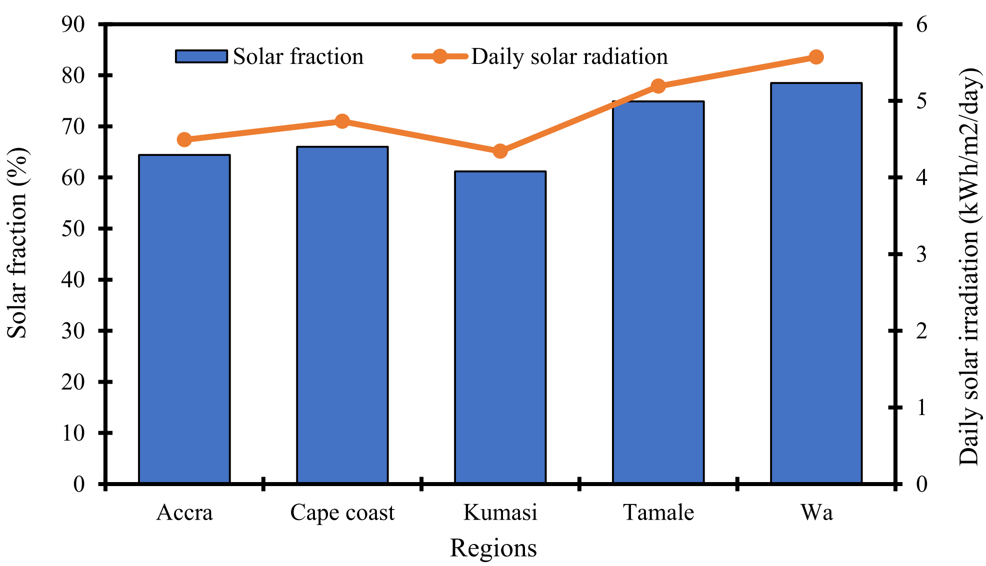


**Fig. 5.** Solar fraction (%) and daily solar irradiation (kWh/m$^2$/day).

of SWH systems' effectiveness in utilising solar energy for heating water in these areas. However, it is essential to note that correlation does not necessarily imply causation. While the findings show a strong association between solar irradiation and SWH system heat delivered, other factors or variables not considered in this analysis could also influence the results.

Fig. 5 shows the estimated solar fraction (SF) from each city's SWH system. The SF is a commonly used solar performance measure that helps identify the fraction of the heat energy required by the solar water heater. It can be seen that, with the same number of solar collectors at each selected location, the SF ranges from 61.2% to 78.5%. The maximum SF is obtained from the SWH system installation at Wa (78.5%). Next are Tamale (74.9%), Cape Coast (66%), Accra (64.4%), and Kumasi, which has a minimum value of 61.2%. This result makes it evident that increasing the amount of solar irradiation can lead to higher heat delivery in these cities. For example, it can be observed that, at Kumasi, where the solar irradiation is the lowest, the SF is also the lowest in the same condition. The SWH is, therefore, feasible at all selected locations, particularly in Wa and Tamale, located in Northern Ghana, which receives higher solar radiation throughout the year. In view of this, the SF obtained for these cities is greater than 50%, which is usually recommended for SWH systems with storage to work efficiently [11]. However, the values could be increased by increasing the number of solar collectors, leading to higher system installation costs [9].

The annual heating energy saved is the difference between the energy that could be generated from a base case (using the utility grid to

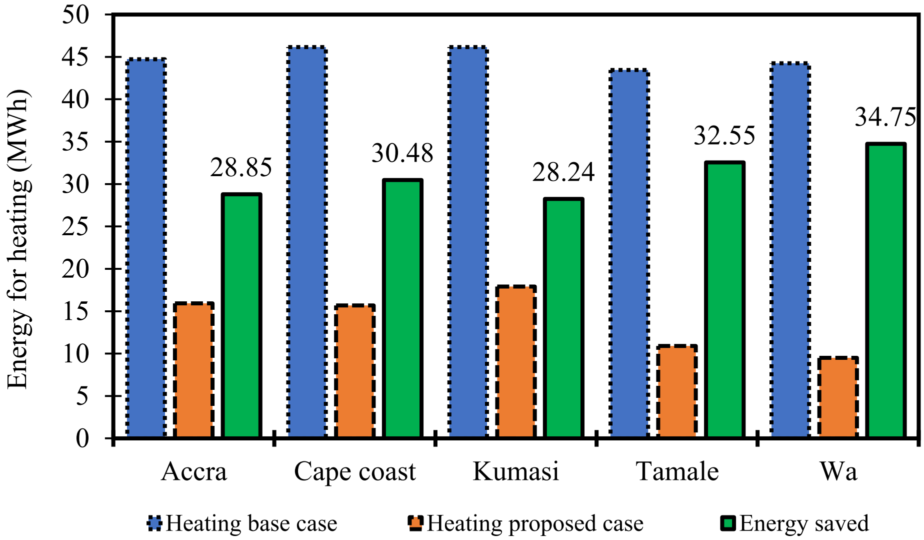


**Fig. 6.** Energy generated for heating and annual energy saved (MWh).

generate heat energy to produce hot water) and the proposed case (using the SWH system to generate heat energy to produce hot water). Singh et al. (2020) alluded to the fact that SF and energy savings are directly related. This implies that a high value of SF would correspond to a high amount of energy being saved in a given city. This is verified here, as it can be observed from Fig. 6, which displays the annual heating energy for different cases. The maximum yearly energy is saved from the system installed in Wa with the highest SF value. The annual energy saved is about 34.75 MWh in Wa, 32.55 MWh in Tamale, 30.48 MWh in Cape Coast, 28.85 MWh in Accra city, and 28.24 MWh in Kumasi.

*GHG emissions analysis*

SWH systems are clean energy technologies and therefore do not contribute to GHG emissions when used for water heating in domestic, commercial, and industrial applications. Thus, the SWH system application is crucial for developing countries willing to switch to renewable energy technologies to reduce emissions. In this study, the SWH system is used in hotels to replace conventional fuel-based systems to estimate GHG emissions reduction potential.

As illustrated in Fig. 7, in the base case where conventional fuel is used for water heating, the GHG emissions are higher compared to the proposed case, which uses solar energy to heat the water. In Accra, for example, installing SWH in a hotel could decrease GHG emissions from 16.9 $tCO_2$ to 6 $tCO_2$ per year, representing a decrease rate of 64.4%. In this case, the annual GHG mitigation potential is estimated at 10.9 $tCO_2$/yr. The maximum GHG mitigation potential is obtained in Wa, with 13.2 $tCO_2$ reduced per year, representing a 78.5% decrease in GHG emissions compared to the base case. The second highest mitigation potential is obtained in Tamale, at 12.7 $tCO_2$/yr, followed by Cape Coast at 11.6 $tCO_2$/yr, Accra at 10.9 $tCO_2$/yr, and Kumasi at 10.7 $tCO_2$/yr.

In addition, the annual emission reduction potential can be illustrated in terms of barrels of crude oil not consumed, as presented in Fig. 7. It is seen that about 30.6 barrels and 24.9 barrels of crude oil could be saved for the SWH system installation in Wa and Kumasi, respectively. Likewise, the emission reduction is equivalent to 28.7 barrels, 26.9 barrels, and 25.4 barrels of crude oil that would not be consumed in Tamale, Cape Coast, and Accra, respectively, if the conventional heating systems are replaced with SWH systems in Ghanaian hotels. These findings are comparable with those obtained by Singh et al. [9] and Soomro et al. [38] in India and Pakistan, respectively, where using SWH in the household sector would considerably reduce GHG emissions.

*Financial analysis*

Fig. 8 displays the SPP and EPP for each city. The SPP considers the total investment and cash inflows, while EPP focuses on the equity investment and equity cash inflows. The EPP indicates how quickly an investor can recover his investment in a project [54]. It can be seen in Fig. 8 that the SPP is higher in Kumasi (6.4 years), followed by Accra (6.3 years), Cape Coast (5.9 years), Tamale (5.5 years), and Wa (5.2 years). This suggests that hotels in these cities can expect to recover their investment in SWH systems within 5 to 6 years. The implication is that the SWH systems can be financially viable over the long term. Besides, the SPP obtained for each city is nearly comparable to the SPP found in other studies by Sinethemba et al. [25] in South Africa, Dioha et al. [30] in Nigeria, Naveed Arif et al. [24] in Pakistan, Singh et al. [9] and Raza et al. [29] in India, Yılmaz [60] in Turkey, and Almahmoud et al. [36] in Saudi Arabia. Moreover, the results indicate that SPP decreases with the increase in solar fraction, with the shortest SPP obtained in Wa, where the SF is the highest.

In addition, the EPP is seen as a better time indicator than the SPP. It can be observed in Fig. 8 that the EPP range from 4.9 to 6.1 years. The EPP period is generally shorter than the SPP, indicating that including the cost of equity capital improves the financial feasibility of the SWH systems. Furthermore, it is noticed that incentives can help reduce the EPP of the SWH system and allow stakeholders to recover their investments (Table 5). More importantly, the NPV is positive for all selected cities, as shown in Table 5, indicating that the project is viable for investment at the different locations. A comparison of EPP and NPV in Table 5 shows that a shorter EPP leads to higher NPV values. For example, an EPP of 6 years in Accra leads to an NPV of USD 8410, while an EPP of 4.9 years in Wa leads to an NPV of USD 14,319. This is the case in all cities, meaning that SWH systems of the same size are more viable in Wa than in Kumasi. This can be explained by the amount of solar irradiation received in each location. According to Quansah et al. [61], solar radiation in Kumasi is lower than in the northern part of Ghana, and this could lead to a higher energy cost.

The study further investigated the impact of initial cost on NPV and EPP, as presented in Table 5. It can be observed that even with an increase of 12.5% and 25% in the initial cost, the NPV remained positive. The positive NPV values indicate that the present value of the projected cash flows associated with the SWH system exceeds the initial investment costs, considering the time value of money. This implies that over the project's lifespan, the anticipated returns and savings from adopting

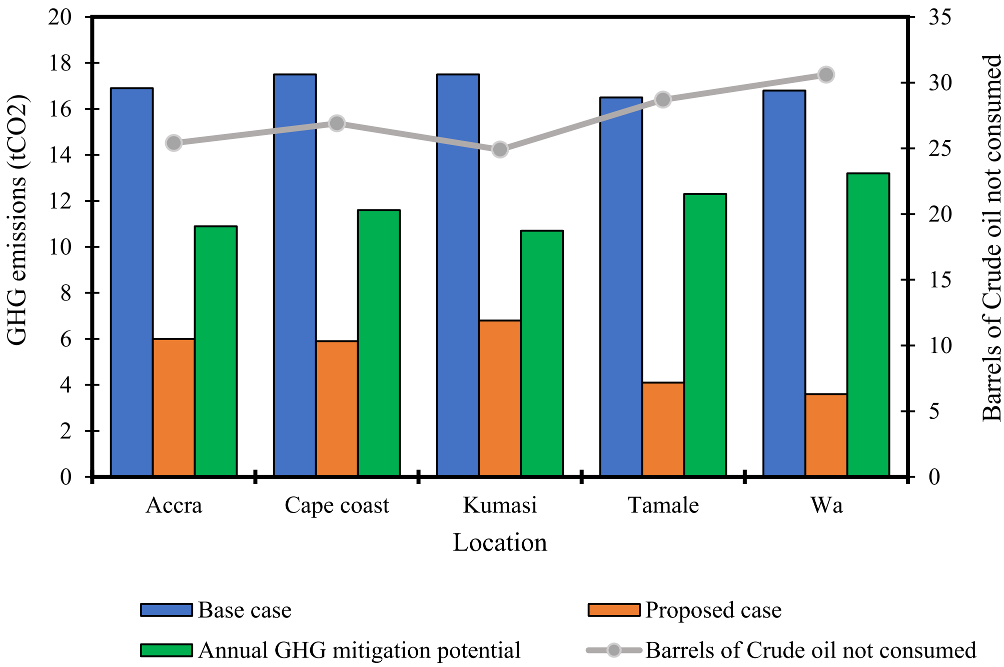


**Fig. 7.** Annual GHG mitigation potential.

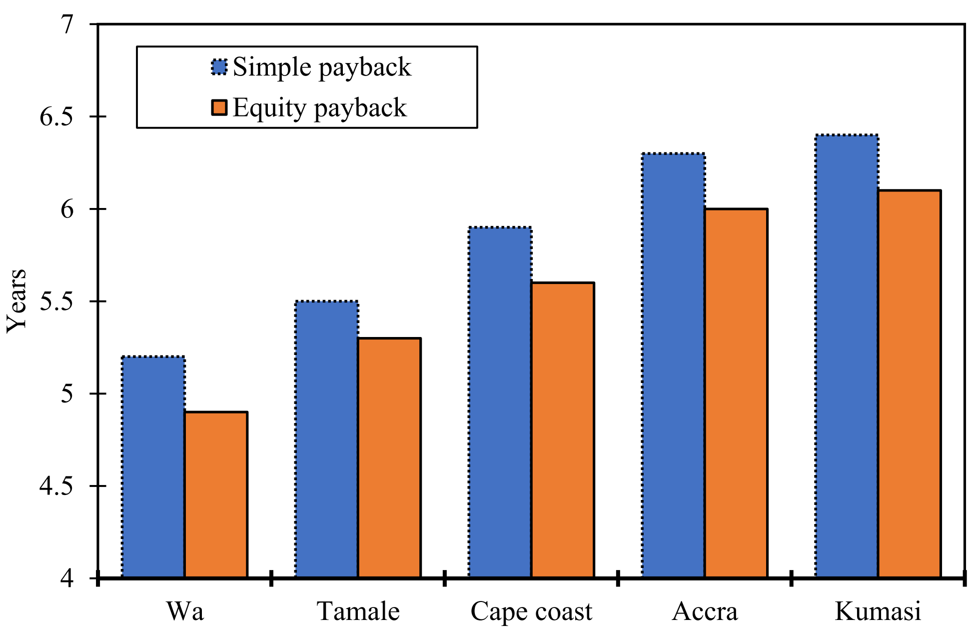


**Fig. 8.** SWH systems simple payback period and equity payback period.

**Table 5**
Sensitivity analysis on initial cost.

| Sensitivity range (%) | Accra | | Cape Coast | | Kumasi | | Tamale | | Wa | |
|---|---|---|---|---|---|---|---|---|---|---|
| | EPP (year) | NPV (USD) | EPP (year) | NPV (USD) | EPP (year) | NPV (USD) | EPP (year) | NPV (USD) | EPP (year) | NPV (USD) |
| −25.5 | 4.5 | 12,710 | 4.3 | 14,379 | 4.6 | 12,155 | 4 | 16,437 | 3.7 | 18,619 |
| −12.5 | 5.3 | 10,560 | 5 | 12,229 | 5.4 | 10,005 | 4.6 | 14,287 | 4.3 | 16,469 |
| **0.0** | **6** | **8410** | **5.6** | **10,079** | **6.1** | **7855** | **5.3** | **12,137** | **4.9** | **14,319** |
| +12.5 | 6.7 | 6260 | 6.3 | 7929 | 6.8 | 5705 | 5.9 | 9987 | 5.5 | 12,169 |
| +25.0% | 7.4 | 4110 | 7 | 5779 | 7.6 | 3555 | 6.5 | 7837 | 6.1 | 10,019 |

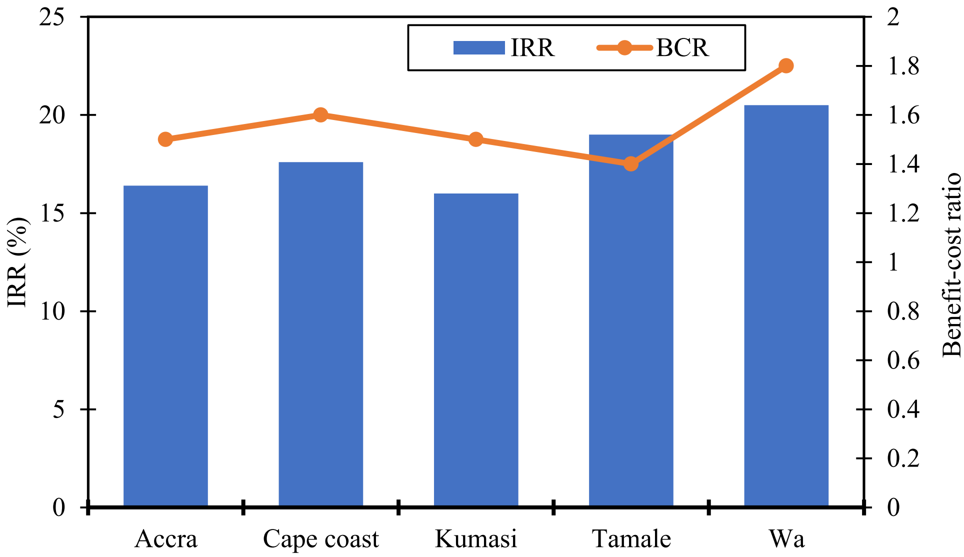


**Fig. 9.** Potential IRR and BCR from SWH system installation at selected cities.

the SWH system will surpass the expenses incurred during its installation.

Correspondingly, the EPP experienced a slight increment with the rise in the initial cost. These outcomes strongly affirm the economic feasibility of implementing the SWH system in Ghanaian hotels. In view of this, the slight increase in the EPP with higher initial costs suggests that it may take a marginally longer time for the cumulative cash inflows to equal the initial equity investment. Despite this slight increase, the EPP remains within an acceptable range, indicating a favourable economic viability of the SWH system in Ghanaian hotel settings.

Fig. 9 illustrates the SWH system's BCR and IRR for each city. The BCR and IRR are also important financial metrics that help in the financial viability analysis of a system. Generally, a BCR greater than 1 indicates that the project is viable (the benefits outweigh the costs), and a BCR less than 1 shows that the project is not attractive for investment. It can be seen in Fig. 9 that the BCR is greater than 1 in all selected cities, ranging from 1.8 in Wa to 1.5 in Kumasi. The implication is that installing SWH systems in hotels in Ghana would provide positive economic outcomes. These systems have the potential to generate benefits that exceed the initial investment costs, making them financially advantageous for hotel owners. Furthermore, as shown in Fig. 9, the SWH system IRR ranges from 16% to 20.5% across Accra, Cape Coast, Kumasi, Tamale, and Wa. These values indicate the profitability and financial attractiveness of investing in SWH systems for hotels in Ghana. A higher IRR implies a greater return on investment, suggesting that installing SWH systems in these cities can be financially beneficial. The IRR values reported here indicate that the systems have the potential to generate satisfactory returns over their operational lifespan.

### *Barriers and recommendations*

Adopting SWH systems in developing countries faces various challenges that hinder its widespread use [62]. One of the significant obstacles is the high initial cost of purchasing and installing the systems, which can be a major deterrent for small businesses and households with limited budgets [35]. Also, inadequate technical expertise and knowledge to design, install, and maintain SWH systems further limit their uptake. In addition, low-quality products and unreliable suppliers also decrease consumer confidence in the technology. The lack of appropriate policy and regulatory frameworks is another significant barrier to adopting SWH systems in developing countries [62]. Likewise, the absence of clear regulations and standards to ensure the safety, quality, and performance of SWH products and installations can lead to the proliferation of substandard products and installations that undermine public trust in the technology.

Despite the abovementioned barriers, stakeholders could benefit from several opportunities by adopting SWH systems. For example, installing SWH systems can provide substantial savings on electricity expenses. In this context, hotels can reduce their reliance on traditional heating methods, resulting in lower energy bills, by utilising the abundant solar energy available. This financial benefit can be particularly appealing to hoteliers, as it directly impacts their bottom line and can contribute to higher profitability.

Notwithstanding the high initial cost of installing the SWH system, it is essential to emphasise the long-term investment aspect. Although payback periods may be longer than traditional heating systems, once the system is fully paid off, the hotel will continue to benefit from free energy from the sun. This can lead to significant savings in the long run, offsetting the initial investment and providing reliable energy at a lower cost in the future. In addition, SWH systems can be designed with backup systems to ensure reliable energy for hotels. For instance, a hybrid system can combine solar energy with other heating methods, such as electricity or gas, to provide uninterrupted service even during low solar irradiation or system maintenance periods. Therefore, by implementing appropriate system design and backup measures, hotels can mitigate concerns about reliability.

Furthermore, though sustainability may not be the primary concern for hoteliers, it is worth highlighting the marketing value of sustainability initiatives, such as "sustainable tourism" or "eco-tourism." Several travellers are increasingly conscious of their environmental impact and actively seek eco-friendly accommodations. Therefore, hotels can enhance their brand image and attract environmentally conscious guests by installing SWH systems. This can increase bookings, customer loyalty, and positive word-of-mouth recommendations. Moreover, providing more national and local budget sovereignty in West Africa for SWH projects can create jobs and value in the region [63].

Policymakers can adopt several measures to promote the growth of solar energy deployment in Africa. In this regard, the following policy changes can help create a conducive environment for the widespread deployment of SWH systems in Ghana:

- The government can introduce preferential policies to encourage the establishment and growth of local solar water heater manufacturing facilities. This can include tax incentives, grants, or low-interest loans to support the setup of assembly plants or manufacturing units. Hence, promoting local production could reduce the cost of solar water heaters, making them more affordable for local and regional markets.

- Targeted awareness campaigns could be launched to educate households, real estate developers, hotels, and commercial facilities about the benefits of SWH systems. These campaigns can highlight the cost savings, environmental advantages, and long-term benefits of using solar energy for water heating. Additionally, the campaigns can provide information about available incentives and support for adopting solar water heaters.
- Develop regulations that require existing commercial properties, especially hotels and large-scale facilities, to install SWH systems. These regulations can be implemented gradually, giving businesses a reasonable compliance timeline. Providing financial incentives or tax breaks for early adopters can further encourage compliance and accelerate the adoption of solar water heaters in commercial buildings.
- Alongside SWH systems, hybrid water heating systems that combine solar energy with electricity should be encouraged. These systems can provide hot water even during periods of limited sunlight. Given this, consumers can have a reliable hot water supply while benefiting from the cost savings and environmental advantages of solar energy-promoting hybrid systems.
- Incorporate solar water heaters into the National Building Code to include provisions that require the installation of solar water heaters in new construction or major renovations. This will ensure that SWH systems become a standard feature in new buildings, contributing to a sustainable and energy-efficient built environment.
- Adjust import duties to make SWH systems more affordable. The government can consider removing import duties on the components required for assembly or manufacturing. This will lower the overall cost of solar water heaters, making them more accessible to consumers. Conversely, imposing higher import duties on electric water heaters can create a price disparity that incentivizes adopting SWH systems.

## Conclusion

This study conducted technical, financial, and emission analyses of adopting SWH systems for hotels in five cities in Ghana: Accra, Cape Coast, Kumasi, Tamale, and Wa. The major new and important outcomes of this study are as follows:

- With a SWH system of the same capacity in the selected cities, the SF ranges from 61.2% to 78.5%. The maximum SF is obtained in Wa (78.5%), which receives the highest solar irradiation, followed by Tamale (74.9%), Cape Coast (66%), Accra (64.4%), and Kumasi (61.2%). All the selected locations have a SF value greater than 50%, which is suitable for SWH systems with storage.
- There is a significant potential for energy savings from deploying the SWH system in hotels in Ghana. About 154.87 MWh of energy could be saved by installing SWH systems instead of conventional water heating systems. Energy savings increase with the increase in solar irradiation, with Wa having the highest energy savings (34.75 MWh).
- A total of 58.7 $tCO_2$ could be avoided annually by installing SWH in the selected cities. The $CO_2$ emissions savings are equivalent to 136.5 barrels of crude oil not consumed annually in the country.
- The NPV is positive for the SWH systems at the selected sites, meaning the project is financially viable and attractive for investment. In addition, the SPP and EPP of the project range from 6.4 to 5.2 years and 6.1 to 4.9 years, respectively.
- The benefit-cost ratio varies between 1.5 and 1.8. Furthermore, incentives would further increase the BCR, making the project more attractive for investment in the sector.

The study findings have revealed that installing SWH systems is feasible in Ghanaian hotels and financially viable for investment. Also, the study findings strongly encourage hoteliers to integrate SWH systems into their operations. The financial viability, reasonable payback periods, environmental benefits, long-term cost savings, and support for sustainability objectives make SWH an appealing option for hotels seeking to reduce energy costs and minimise their environmental impact. Therefore, appropriate policy and regulatory frameworks must be developed to ensure the quality of SWH products and installations. Financial incentives such as tax credits and subsidies are needed to make SWH systems more accessible and affordable for hotels and small businesses in Ghana.

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## Declaring Conflicts of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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